

Module-1: INTRODUCTION TO DATA MINING

Data mining, Knowledge Discovery process, Data Mining Functionalities-Kinds of Patterns, Major Issues in Data Mining, Data Objects and Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Dissimilarity of Numeric Data-Distance measures, Data Pre-processing: Major Tasks in Data Pre-processing, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization.

1 INTRODUCTION TO DATA MINING

- 1.1 Data Mining
- 1.2 Knowledge Discovery Process
- 1.3 Data Mining Functionalities-Kinds of Patterns
- 1.4 Major Issues In Data Mining
- 1.5 Data Objects and Attribute Types
- 1.6 Basic Statistical Descriptions of Data
- 1.7 Data Visualization
- 1.8 Dissimilarity of Numeric Data-Distance measures

DATA PREPROCESSING

- 1.9 Major Tasks in Data Preprocessing
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1.1 Data Mining

Data mining (knowledge discovery in databases):

- *Extraction or mining of interesting (implicit, previously unknown and potentially useful) information / knowledge or patterns from large amounts of databases.*
- *Process of analyzing data from different perspectives, summarizing it into useful information that can be used to increase revenue, cuts costs or both.*

Alternative names:

- Knowledge Discovery (mining) in Databases (KDD)
- knowledge extraction
- data/pattern analysis
- data dredging
- information harvesting
- business intelligence, etc.

What is not data mining?

- Not (Deductive) query processing.
- Not Expert systems or statistical programs

Data Mining — Potential Applications

Where are the data sources for analysis?

- Credit card transactions, loyalty cards, discount coupons, customer complaint calls, plus (public) lifestyle studies
- **Target marketing:** Find clusters of “model” customers who share the same characteristics: interest, income level, spending habits, etc.
- **Determine customer purchasing patterns over time:** Conversion of single to a joint bank account: marriage, etc.
- **Cross-market analysis:** Associations/co-relations between product sales. Prediction based on the association information
- **Customer profiling:** data mining can tell you what types of customers buy what products (clustering or classification)
- **Identifying customer requirements:** identifying the best products for different customers, use prediction to find what factors will attract new customers
- **Provides summary information:** various multidimensional summary reports, statistical summary information (data central tendency and variation)
- **Corporate Analysis and Risk Management**
- Finance planning and asset evaluation
 - cash flow analysis and prediction
 - contingent claim analysis to evaluate assets
 - cross-sectional and time series analysis (financial-ratio, trend analysis, etc.)
- **Resource planning:**
 - summarize and compare the resources and spending

➤ **Competition:**

- monitor competitors and market directions
- group customers into classes and a class-based pricing procedure

➤ set pricing strategy in a highly competitive market

➤ **Fraud Detection and Management**

Approach: use historical data to build models of fraudulent behavior and use data mining to help identify similar instances

Examples

- Auto insurance: detect a group of people who stage accidents to collect on insurance
- Money laundering: detect suspicious money transactions
- Medical insurance: detect professional patients and ring of doctors and ring of references

➤ **Detecting inappropriate medical treatment**

- Australian Health Insurance Commission identifies that in many cases blanket screening tests were requested (save Australian \$1m/yr).

➤ **Detecting telephone fraud**

- Telephone call model: destination of the call, duration, time of day or week. Analyze patterns that deviate from an expected norm.

Database Processing Vs.	Data Mining Processing
Query – Well defined – SQL	Query – Poorly defined – No precise query language
Data – Operational / Transactional / Current data	Data – Not operational data / Historical Data
Output – Precise – Subset of database	Output – Fuzzy – Not a subset of database
Query Examples – Find all credit card applicants with last name of Ramesh.	Query Examples – Find all credit card applicants who are poor credit risks. (Classification)
– Find all customers who have purchased Data Mining Book	– Find all books which are frequently purchased with Data Mining book. (Association rules)
– Identify customers who have purchased more than Rs10,000 in the last month.	– Identify customers with similar buying habits. (Clustering)

1.2 Knowledge Discovery in Databases (KDD) Process

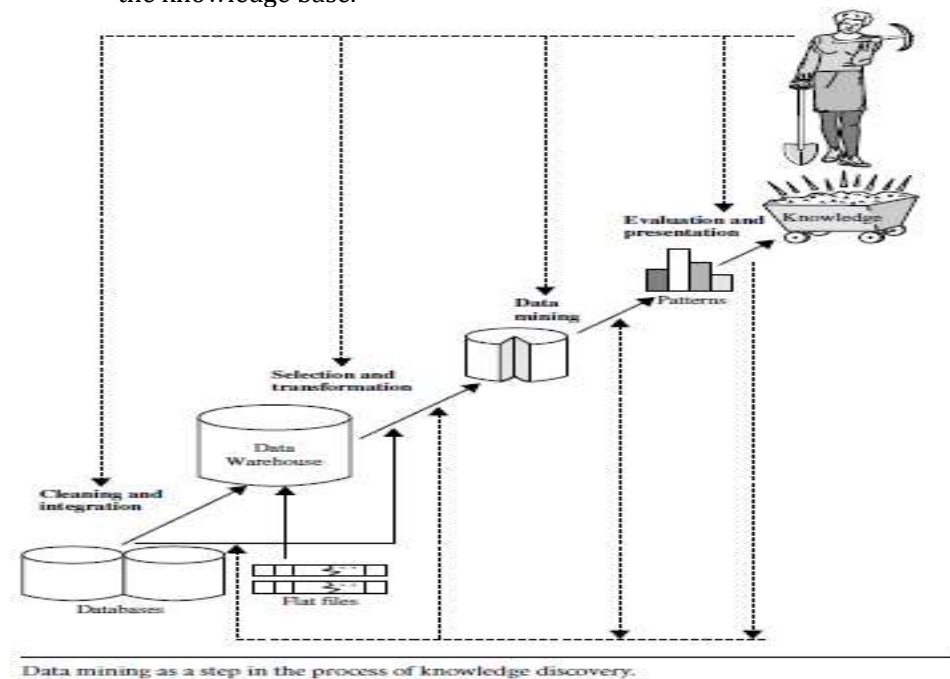
The KDD process is shown in Figure 1.2(a) as an iterative sequence of the following steps:

- 1) Data cleaning
 - 2) Data integration
 - 3) Data selection
 - 4) Data transformation
 - 5) Data mining
 - 6) Pattern evaluation
 - 7) Knowledge presentation
- The ultimate goal is to extract high-level knowledge from low-level data.
 - Major KDD application areas include:
 - marketing, fraud detection, telecommunication and manufacturing.

Steps of a KDD Process

Firstly identify the goal of the KDD process from the customer's perspective. Understand application domains and relevant prior knowledge. Then select a target data set or subset of data samples on which discovery is to be performed.

- 1) Data cleaning to remove noise and inconsistent data.
- 2) Data integration where multiple data sources may be combined.
- 3) Data selection where data relevant to the analysis task are retrieved from the database.
- 4) Data transformation where data are transformed (converted) and consolidated into new and common format(s) appropriate for mining.
 - Steps 1 through 4 are different forms of data preprocessing, where data are prepared for mining.
- 5) Data mining an essential process where intelligent methods are applied to extract data patterns or desired results.
 - The data mining step may interact with the user or a knowledge base.
- 6) Pattern evaluation to identify the truly interesting patterns representing knowledge based on interestingness measures.
- 7) Knowledge presentation where visualization and knowledge representation techniques are used to present mined knowledge (results) to users in meaningful manner.
 - The interesting patterns are presented to the user and may be stored as new knowledge in the knowledge base.



Kinds of Data Can Be Mined

Data mining can be applied to any kind of data as long as the data are meaningful for a target application. The most basic forms of data for mining applications are:

Database-oriented data sets and applications:

- 1) Relational database (Records of Customers, Items): Database Management System (DBMS), RDBMS, Entity-Relationship (ER) data model.
- 2) Data warehouses (multiple sources data): A Data Warehouse *is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.*
- 3) Transactional database (transaction of customer's purchase, web clicks): *Each record in a Transactional Database captures a transaction, which typically includes a unique transaction identity number (TransID) and a list of the items making up the transaction such as a customer's purchase, a flight booking, or a user's clicks on a web page in the transaction.*

Advanced sets and advanced applications: There are many other kinds of data that have versatile forms and structures and rather different semantic meanings.

These applications bring about new challenges, like how to handle data carrying special structures (e.g., sequences, trees, graphs, and networks) and specific semantics (such as ordering, image, audio and video contents, and connectivity), and how to mine patterns that carry rich structures and semantics.

- 4) Object-Relational Database Systems (complex object structures, class and subclass hierarchies)
- 5) Temporal, Sequence and Time-series database systems (historical records, stock exchange and biological sequence data)
- 6) Spatial and Spatiotemporal contain *spatial-related information*. Examples include geographic (map) databases, very large-scale integration (VLSI) or Computed-Aided Design databases, and medical and satellite image databases.
- 7) Hypertext and Multimedia database systems (text, image, audio and video data)
- 8) Heterogeneous and Legacy database systems (Multisource data)
- 9) Data Streams management systems (video surveillance and sensor data, which are continuously transmitted)
- 10) World Wide Web -based global information systems.

1.3 Data Mining Functionalities (Tasks)

1) Characterization and Discrimination

- *Descriptions of a class or a concept* are called Class/Concept Descriptions.
- These descriptions can be derived via:

i. Data Characterization *is a summarization of the general characteristics or features of a target class of data.*

Output forms: pie charts, bar charts, curves, multidimensional data cubes, and multidimensional tables, including crosstabs.

ii. Data Discrimination *is a comparison of the general features of target class data objects with the general features of objects from one or a set of contrasting classes.*

Output forms: similar to characteristic descriptions.

The Discrimination descriptions expressed in rule form are referred to as discriminant rules.

2) Mining Frequent Patterns

- Frequent patterns *are patterns that occur frequently in data.*
Kinds: itemsets, subsequences, and substructures.
- A Frequent Itemset *is a set of items that frequently appear together in a transactional data set,* such as milk and bread.
- A Frequent Subsequence *is a sequence of patterns that occur frequently such as purchasing a PC, followed by a camera, and then a memory card.*
- A substructure *can refer to different structural forms, such as graphs, trees, or lattices, which may be combined with itemsets or subsequences.*
- If a substructure occurs frequently, it is called a (frequent) structured pattern.
- ❖ Mining frequent patterns leads to the discovery of interesting associations and correlations within data.

3) Associations, and Correlations

- Association *is a process of uncovering the relationship among data and determining association rules.*
- Association rules that contain a single predicate are referred to as single-dimensional association rules.
- Buys(X, “computer”) => buys(X, “software”) [support = 1%, confidence = 50%]
Can be written as “computer => software [1%, 50%]”
- Adopting the terminology used in multidimensional databases, where each attribute is referred to as a dimension, the above rule can be referred to as a multidimensional association rule.
- Mining of Correlations *is a kind of additional analysis performed to uncover interesting statistical correlations between associated-attribute-value pairs or between two item sets to analyze that if they have positive, negative or no effect on each other.*

4) Classification and Prediction for Predictive Analysis

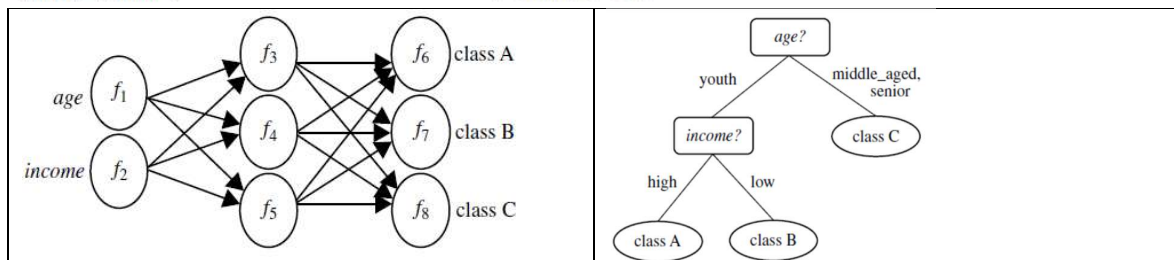
- There are two forms of data analysis that can be used for extracting models describing important classes or to predict future data trends. These two forms are as follows –
- Classification
- Prediction
- Classification *is the process of finding a model (or function) that describes and distinguishes data classes or concepts.*
 - A model or classifier is constructed to predict categorical labels whose class label is unknown.
 - The derived model is based on the analysis of a set of training data (i.e., data objects whose class label is known).
- Ex: classify animals based on (spine), or classify cars based on (fuel mileage)
- A classification model can be represented in various forms, such as (a) classification (IF-THEN) rules, (b) a decision tree, or a (c) neural network.

$age(X, \text{“youth”}) \text{ AND } income(X, \text{“high”}) \longrightarrow class(X, \text{“A”})$

$age(X, \text{“youth”}) \text{ AND } income(X, \text{“low”}) \longrightarrow class(X, \text{“B”})$

$age(X, \text{“middle_aged”}) \longrightarrow class(X, \text{“C”})$

$age(X, \text{“senior”}) \longrightarrow class(X, \text{“C”})$



- A Decision Tree is a flow-chart-like tree structure, where each internal node (non-leaf node) denotes a test on an attribute value, each branch represents an outcome of the test, and each leaf nodes (or terminal node) represent classes or class distributions. Decision trees can easily be converted to classification rules.
- A Neural Network, when used for classification, is typically a collection of neuron-like processing units with weighted connections between the units.
- There are many other methods for constructing classification models, such as Naive Bayesian classification, Support Vector Machines, and k-Nearest Neighbor Classification.
- The resulting classification should maximally distinguish each class from the others, presenting an organized picture of the data set.
- Prediction models predicts continuous-valued functions or ordered value as opposed to a categorical label. and refer to both numeric prediction and class label prediction
- Regression analysis is a statistical methodology that is most often used for numeric prediction
- Classification and prediction may need to be preceded by relevance analysis, which attempts to identify attributes that do not contribute to the classification or prediction process. These attributes can then be excluded.

5) Cluster Analysis

- Cluster is a group of objects that belongs to the same class. Similar objects are grouped in one cluster and dissimilar objects are grouped in another cluster.
- Clustering is the process of making a group of abstract objects into classes of similar objects.
- Ex: market research, pattern recognition, data analysis, and image processing
- The objects are clustered or grouped based on the principle of maximizing the intraclass similarity and minimizing the interclass similarity.
- Each formed cluster can be viewed as a class of objects, from which rules can be derived. Clustering can also facilitate taxonomy formation, that is, the organization of observations into a hierarchy of classes that group similar events together.

6) Outlier Analysis and Evolution Analysis

- Outliers are the data objects of a database that do not comply with the general behavior or model of the data. Most data mining methods discard outliers as noise or exceptions.
 - Ex: Useful in fraud detection, rare events analysis.
- Rather than using statistical or distance measures, deviation-based methods identify outliers by examining differences in the main characteristics of objects in a group.
- Data Evolution Analysis describes and models regularities or trends for objects whose behavior changes over time.
 - Trend and deviation: Ex:-regression analysis
 - Sequential pattern mining: Ex-digital camera -> large SD memory
 - Periodicity analysis
 - Similarity-based analysis

Kinds of Patterns

➤ Data mining system can mine multiple kinds of patterns to accommodate different user expectations or applications and should be able to discover patterns at different levels of abstraction.

➤ In general, data mining tasks can be classified into three categories:

- I. Descriptive Analytics: insight into the past
- II. Predictive Analytics: understanding the future
- III. Prescriptive Analytics: advise on possible outcomes

I. **Descriptive mining model (Business Intelligence)** *characterize the general properties of the data in the database.* Use data aggregation and data mining to provide insight into the past and answer: “What has happened?”

- Class/Concept Description: Characterization and Discrimination
- Mining Frequent Patterns
- Associations
- Correlations
- Cluster Analysis
- Outlier Analysis
- Evolution Analysis

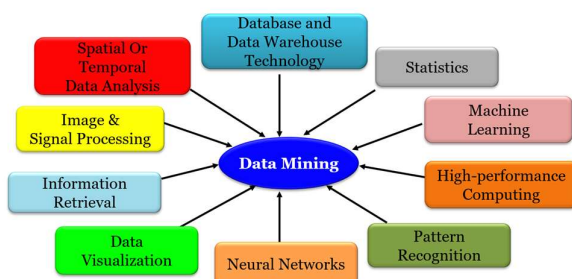
II. **Predictive mining model (Forecasting)** *perform inference on the current data in order to make predictions.* Predictive analytics turns data into valuable, actionable information which use statistical models and forecasts techniques to understand the future and answer: “What could happen?”

- Classification
- Regression
- Time series Analysis
- Prediction

III. **Prescriptive (Optimization and Simulation)** analytics goes beyond predicting future outcomes by also *suggesting actions to benefit from the predictions.* Prescriptive analytics not only *anticipates what will happen and when it will happen, but also why it will happen.* which use optimization and simulation algorithms to advice on possible outcomes and answer: “What should we do?”

Data Mining Technologies

As a highly application-driven domain, data mining is an interdisciplinary field, the confluence of a set of disciplines and has incorporated many techniques from other domains as shown below. The interdisciplinary nature of data mining research and development contributes significantly to the success of data mining and its extensive applications.



1.4 Major Issues in Data Mining

- Data mining is a dynamic and fast-expanding field with great strengths. The major issues in data mining research are partitioned them into five groups:
 - 1) Mining Methodology
 - 2) User Interaction
 - 3) Efficiency and scalability
 - 4) Diversity of Database types
 - 5) Data mining and society

1) Issues related to Mining Methodology:

Researchers have been vigorously developing new data mining methodologies. This involves:

- Mining various and new kinds of knowledge: Data mining covers a wide spectrum of data analysis and knowledge discovery tasks, from data characterization and discrimination to outlier analysis. These tasks may use the same database in different ways and require the development of numerous data mining techniques.
- Data Mining knowledge in multidimensional space: Data can be aggregated or viewed as a multidimensional data cube, by combinations of dimensions (attributes) at varying levels of abstraction.
- Data mining—an interdisciplinary effort: The power of data mining can be substantially enhanced by integrating new methods from multiple disciplines like information retrieval, natural language processing, software engineering knowledge into the data mining process.
- Boosting the power of discovery in a networked environment: Knowledge derived in one set of objects can be used to boost the discovery of knowledge in a “related” or semantically linked set of objects resided in a linked or interconnected environment.
- Handling uncertainty, noise, or incompleteness of data: Data cleaning, data preprocessing, outlier detection and removal, and uncertainty reasoning are examples of techniques that need to be integrated with the data mining process.
- Pattern evaluation and pattern- or constraint-guided mining: Techniques are needed to assess the interestingness of discovered patterns based on subjective measures with respect to a given user class.

2) Issues related to User Interaction:

The user plays an important role in the data mining process. Interesting areas of research include:

- Interactive mining: Interactive mining should allow users to dynamically change the focus of a search, to refine mining requests based on returned results, and to drill, dice, and pivot through the data and knowledge space interactively, dynamically exploring “cube space” while mining.
- Incorporation of background knowledge: Background knowledge, constraints, rules, and other information regarding the domain under study should be incorporated into the knowledge discovery process for pattern evaluation as well as to guide the search toward interesting patterns.

- Ad hoc data mining and data mining query languages: High-level data mining query languages or other high-level flexible user interfaces will give users the freedom to define ad hoc data mining tasks.
- Presentation and visualization of data mining results: Data mining system should adopt expressive knowledge representations, user-friendly interfaces, and visualization techniques.

3) Issues related to Efficiency and scalability are always considered when comparing data mining algorithms.

- Efficiency and scalability of data mining algorithms: Efficiency, scalability, performance, optimization, and the ability to execute in real time are key criteria that drive the development of many new data mining algorithms, to effectively extract information from huge amounts of data.
- Parallel, distributed, and incremental mining algorithms: Cloud computing and cluster computing, which use computers in a distributed and collaborative way to tackle very large-scale computational tasks, are also active research themes in parallel data mining.

4) Issues related to Diversity of Database types include:

- Handling complex types of data: Domain- or application-dedicated data mining systems are being constructed for in depth mining of specific kinds of data. The construction of effective and efficient data mining tools for diverse applications remains a challenging and active area of research.
- Mining dynamic, networked, and global data repositories: Web mining, multisource data mining, and information network mining have become challenging and fast-evolving data mining fields.

5) Issues related to Data mining and society are

- Social impacts of data mining: How can we use data mining technology to benefit society? How can we guard against its misuse? The improper disclosure or use of data and the potential violation of individual privacy and data protection rights are areas of concern that need to be addressed.
- Privacy-preserving data mining: Data mining will help scientific discovery, business management, economy recovery, and security protection (e.g., the real-time discovery of intruders and cyberattacks).
- Invisible data mining: More and more systems should have data mining functions built within so that people can perform data mining or use data mining result simply by mouse clicking, without any knowledge of data mining algorithms. Users may be unaware that the Intelligent search engines and Internet-based stores is likely collecting data on the buying patterns of its customers, which may be used to recommend other items for purchase in the future.

1.5 Data Objects and Attribute Types

- Data sets are made up of data objects.
- A Data Object represents an entity. In a sales database, the objects may be customers, items and sales.
- Data objects are typically described by attributes. Data objects can also be referred to as samples, examples, instances, data points, or objects.
- If the data objects are stored in a database, they are Data Tuples. That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.
- An Attribute is a data field, representing a characteristic or feature of a data object. The nouns attribute (used in data mining), dimension (used in data warehousing), feature (in Machine learning), and variable (in statistics) are often used interchangeably.

Ex: Customer_ID, Name, and Address etc. are the attributes describing a Customer object

- Observed values for a given attribute are known as Observations.
- An Attribute Vector (or feature vector) is a set of attributes used to describe a given object.
- A Univariate Distribution of data involves one attribute (or variable).
- A Bivariate Distribution involves two attributes, and so on.

The type of an attribute is determined by the set of possible values like:

- 1) Nominal Attribute
- 2) Binary Attribute
- 3) Ordinal Attribute
- 4) Numeric Attribute
- 5) Discrete & Continuous Attributes

1) The values of a **Nominal Attribute** (relating to names) are symbols or names of things that represents some kind of category, code, or state, and so referred to as Categorical and also as Enumerations.

Ex: *hair_color - black, brown, white; marital_status - single, married, divorced, and widowed of person* objects.

As nominal attribute values do not have any meaningful order about them and are not quantitative, it makes no sense to find the mean value or median value. But *attribute's most commonly occurring value* known as the mode, is one of the measures of central tendency.

2) A **Binary Attribute** is a nominal attribute with only two categories or states: 0 or 1, where 0 means the attribute is absent, and 1 is present. Binary attributes are referred to as Boolean if the two states correspond to true and false.

Ex: The attribute medical test is binary, where a value of 1 means the result of the test for the patient is positive, while 0 is negative.

- A binary attribute is symmetric if both of its states are equally valuable and carry the same weight; that is, there is no preference on which outcome should be coded as 0 or 1.

Ex: gender having the states male and female.

- A binary attribute is asymmetric if the outcomes of the states are not equally important.

Ex: Medical test for HIV is 1 (e.g., HIV positive) and 0 (negative).

3) An **Ordinal Attribute** is an attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known.

Ex: Size - small, medium, and large.

- ❖ Nominal, binary, and ordinal attributes are qualitative. They describe a feature of an object without giving an actual size or quantity.

4) **A Numeric Attribute** is quantitative (measurable), represented in integer or real values, and can be

- Interval-scaled Attributes are measured on a scale of equal-size units whose order can be positive, 0, or negative.

Ex: outdoor temperature value for a number of different days.

- A Ratio-scaled Attribute is a numeric attribute with an inherent zero-point. That is, a value can be a multiple (or ratio) of another value.

Ex: weight, height, latitude and longitude

- The values are ordered, and we can also compute the difference between values, as well as the mean, median, and mode.

5) **A Discrete Attribute** has a finite or countably infinite set of values, which may or may not be represented as integers.

Ex: natural numbers, Cust_ID are countably infinite.

If an attribute is not discrete, it is Continuous Attributes and are typically represented as floating-point variables.

1.6 Basic Statistical Descriptions of Data

- Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers. 3 areas of basic statistical descriptions are:

1) Measures of central tendency which measure the location of the middle or center of a data distribution like:

- i. Mean
- ii. Median
- iii. Mode
- iv. Midrange.

2) Dispersion of the data is how are the data spread out? The most common data dispersion measures are the

- i. Range
- ii. Quartiles
- iii. Interquartile range
- iv. Five-number summary
- v. Boxplots
- vi. Variance
- vii. Standard deviation

3) Graphic Displays of Basic Statistical Descriptions of Data presentation software packages include:

- i. Bar charts
- ii. Pie charts
- iii. Line graphs,

Summaries and distributions include:

- i. Quantile plots
- ii. Quantile–quantile plots,
- iii. Histograms
- iv. Scatter plots.

1) Measures of Central Tendency:

i. Mean, ii. Median, iii. Mode and iv. Midrange

- Measures of central tendency include the mean, median, mode, and midrange.

i. Mean

- Let $x_1, x_2, \dots, x_N$ be a set of N values or observations, such as for some numeric attribute X , like salary.
- The Arithmetic Mean (Average) of this set of values is obtained by dividing sum of the values of all observations in a series ($\sum X$) by the number of items (N) constituting the series.

$$\text{Mean} = \frac{\text{Sum of the items}}{\text{Number of the items}} = \frac{\sum X}{N} \quad \bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + x_3 + \dots + x_N}{N}$$

(Eq.1)

- This corresponds to the built-in aggregate function, average (avg() in SQL), provided in relational database systems.
- Weighted arithmetic mean (weighted average) is calculated as:

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_N x_N}{w_1 + w_2 + w_3 + \dots + w_N} \quad (\text{Eq.2})$$

- Trimmed mean is the mean obtained after chopping off values at the high and low extremes.

ii. Median

- The median is that value of the variable which divides the group into two equal parts, one part comprising of all values greater, and the other, all values less than median.

$$\text{Median}(M) = \text{Size of } \left(\frac{N+1}{2} \right) \text{th item} \quad (\text{Eq.3})$$

- Calculation of Median – Discrete series:

- Arrange observations in an ordered array.
- If number of terms is odd, the median is the middle term of the ordered array.
- If even, the median is the average of the middle two terms.

- Calculation of median – Continuous series:

$$\text{median} = L_1 + \left(\frac{N/2 - (\sum \text{freq})l}{\text{freq}_{\text{median}}} \right) \text{width} \quad (\text{Eq.4})$$

- where

L_1 -lower boundary of the median interval,

N -number of values in the entire data set,

$(\sum \text{freq})l$ -sum of the frequencies of all of the intervals that are lower than the median interval,

$\text{freq}_{\text{median}}$ -frequency of the median interval, and

width -width of the median interval.

iii. Mode

- The mode for a set of data is the value that occurs most frequently in the set. It can be determined for qualitative and quantitative attributes.
- Data sets with one, two, or three modes are respectively called unimodal, bimodal, and trimodal and data set with two or more modes is multimodal. If each data value occurs only once, then there is no mode.

$$\text{Mean} - \text{Mode} = 3 \times (\text{Mean} - \text{Median})$$

- Empirical relation: (or) $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$

iv. Midrange

- The midrange is the average of the largest and smallest values in the set.

Suppose we have the following values for *salary* (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

$$\text{i) Mean} = \frac{30+36+47+50+52+52+56+60+63+70+70+110}{12} = \frac{696}{12} = 58 \Rightarrow \$58,000$$

$$\text{ii) Median} = \frac{52+56}{2} = \frac{108}{2} = 54; \text{ then median is } \$54,000$$

iii) Mode= Bimodal. The two modes are \$52,000 and \$70,000.

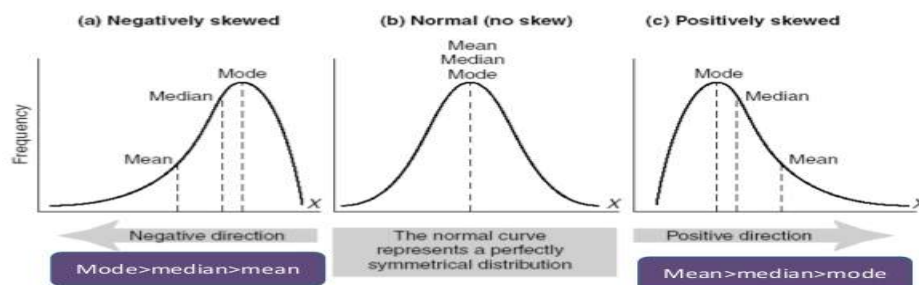
$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean} = 3 \times 54 - 2 \times 58 = 46 \Rightarrow \$46,000$$

$$\text{iv) Midrange} = \frac{30+110}{2} = \frac{140}{2} = 70 \Rightarrow \$70,000$$

Relative positions of Mean Vs Median Vs Mode

- In a unimodal frequency curve with perfect symmetric data distribution, the mean, median, and mode are all at the same center value, as shown in Fig(b).
- Data may be either Positively skewed, where the mode occurs at a value that is smaller than the median (Fig.c), or
- Negatively skewed, where the mode occurs at a value greater than the median (Fig.a).

Position of mean median mode



- From the previous example we analyze that Mean (\$58,000) > Median(\$54,000) > Mode(\$46,000)
Hence we can infer that the data set is Positively Skewed

2) Measuring the Dispersion of Data:

i.

ii.

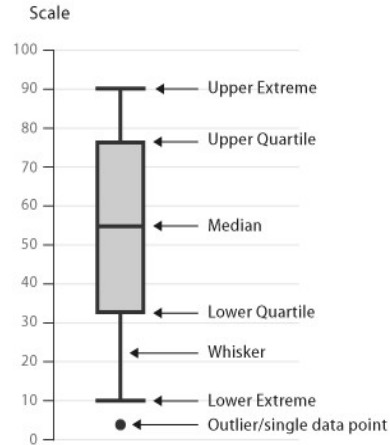
iii.

iv.

v.

vi.

i.



Range,

Quartiles,

Interquartile Range,

Five-number summary,

Boxplots,

Variance,

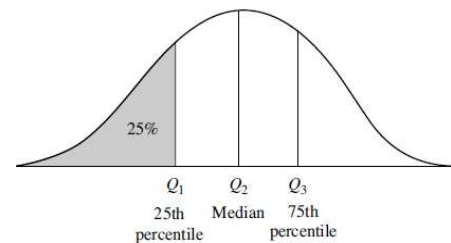
vii. Standard Deviation

➤ Let $x_1, x_2, \dots, x_N$ be a set of observations for some numeric attribute, X which is sorted in increasing numeric order.

Range

➤ The Range of the set is the difference between the largest ($\max()$) and smallest ($\min()$) values.

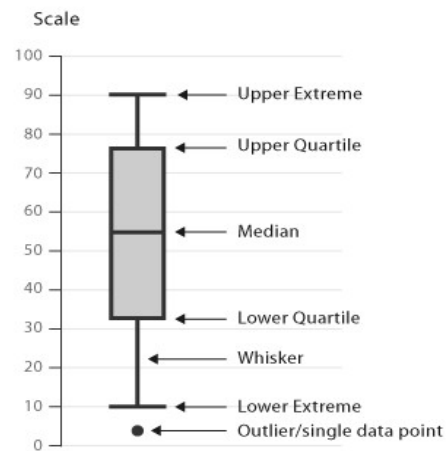
- Quantiles are points taken at regular intervals of a data distribution, dividing it into essentially equal size consecutive sets.
- The 2-quantile is the data point dividing the lower and upper halves of the data distribution. It corresponds to the median.
- The 4-quantiles are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as Quartiles. The 100-quantiles are more commonly referred to as Percentiles; they divide the data distribution into 100 equal-sized consecutive sets.
- The quartiles give an indication of a distribution's center, spread, and shape. The first quartile, denoted by Q_1 , is the 25th percentile. It cuts off the lowest 25% of the data.
- The third quartile, denoted by Q_3 , is the 75th percentile—it cuts off the lowest 75% (or highest 25%) of the data. The second quartile is the 50th percentile. As the median, it gives the center of the data distribution.



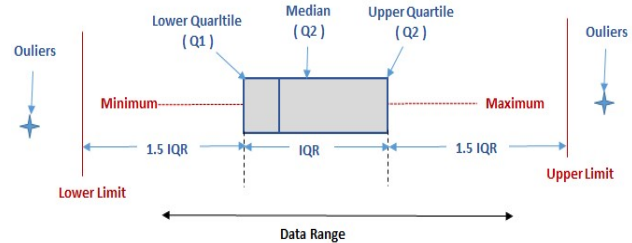
InterQuartile Range (IQR),

Five-number summary, Boxplots

- The distance between the first and third quartiles which is the range covered by the middle half of the data is called the InterQuartile Range (IQR) and is defined as $IQR = Q_3 - Q_1$.
- The five-number summary of a distribution consists of the median (Q_2), the quartiles Q_1 and Q_3 , and the smallest and largest individual observations, written in the order of: Minimum, Q_1 , Median, Q_3 , Maximum.
- Boxplots are a popular way of visualizing a distribution. A boxplot incorporates the five-number summary as follows:



- Typically, the ends of the box are at the quartiles so that the box length is the interquartile range.
- The median is marked by a line within the box.
- Two lines (called whiskers) outside the box extend to the smallest (Minimum) and largest (Maximum) observations.



Variance and Standard Deviation

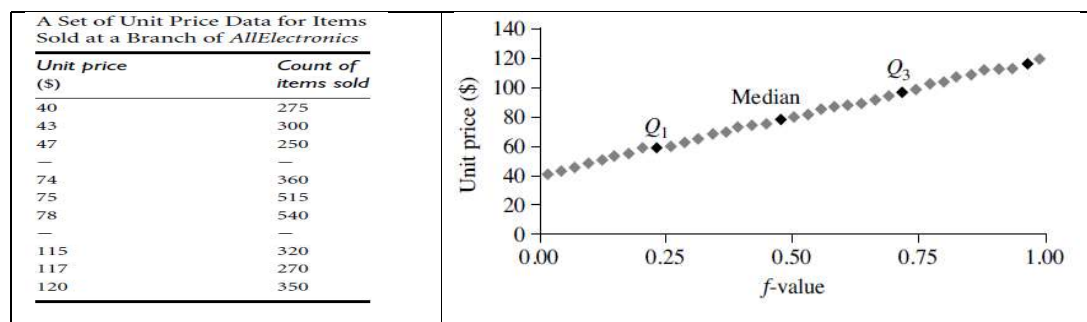
- Variance and standard deviation are measures of data dispersion. They indicate how spread out a data distribution is scalable in large databases.
- A low standard deviation means that the data observations tend to be very close to the mean, while a high standard deviation indicates that the data are spread out over a large range of values.
- The variance of N observations, $x_1, x_2, \dots, x_N$, for a numeric attribute X is:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2 \quad (\text{Eq.4})$$

- Where $\bar{x}$ is the mean value of the observations, as defined in Eq.1.
- The standard deviation of the observations is the square root of the variance.

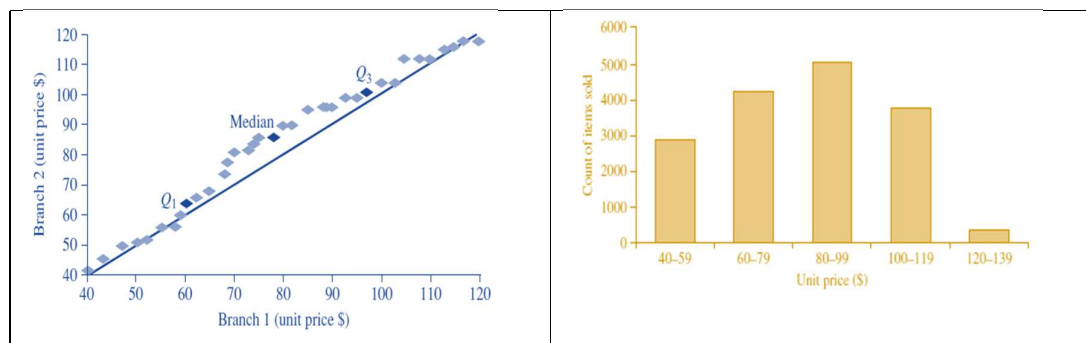
3) Graphic Displays:

- Quantile plots,**
- Quantile-Quantile plots,**
- Histograms, and**
- Scatter plots.**
- Graphs are helpful for the visual inspection of data, which is useful for data preprocessing.
- The quantile plots, quantile-quantile plots, histograms show univariate distributions (i.e., data for one attribute), while scatter plots show bivariate distributions (i.e., involving two attributes).
- A quantile plot is a simple and effective way to have a first look at a univariate data distribution.
- First, it displays all of the data for the given attribute (allowing the user to assess both the overall behavior and unusual occurrences).
- Second, it plots quantile information.



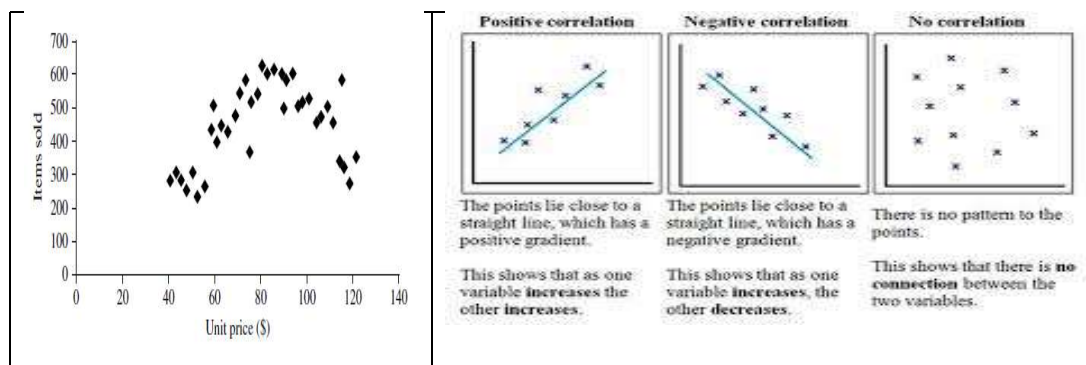
Quantile-Quantile plots

- A quantile-quantile plot, or q-q plot, graphs the quantiles of one univariate distribution against the corresponding quantiles of another.
- From the given q-q plot for unit price data of items sold at two branches during a given time period, we infer that 25% of items sold at branch 1 were less than or equal to \$60, while 25% of items sold at branch 2 were less than or equal to \$64.
- Histograms: “Histos”-pole, and “gram”-chart, histogram-chart of poles. If X is nominal, then a pole or vertical bar is drawn for each known value of X whose height indicates the frequency (i.e., count) of that X value. The resulting graph is more commonly known as a bar chart.
- The range of values for X is partitioned into disjoint consecutive subranges, referred to as buckets or bins whose range is known as width. Typically, the buckets are of equal width.
- For each subrange, a bar is drawn with a height that represents the total count of items observed within the subrange.



Scatter plots

- A scatter plot is one of the most effective graphical methods for *determining* if there appears to be a *relationship, pattern, or trend between two numeric attributes*.
- To construct a scatter plot, each pair of values is treated as a pair of coordinates and plotted as points in the plane.
- The scatter plot is a useful method for providing a first look at bivariate data to see clusters of points and outliers, or to explore the possibility of correlation relationships.
- Two attributes, X, and Y, are correlated if one attribute implies the other. Fig(a), Fig(b) & Fig(c) shows examples of positive correlation, negative correlation and no correlation relationship between two attributes in each of the given data sets.. A line of best fit can be drawn to study the correlation between the variables.

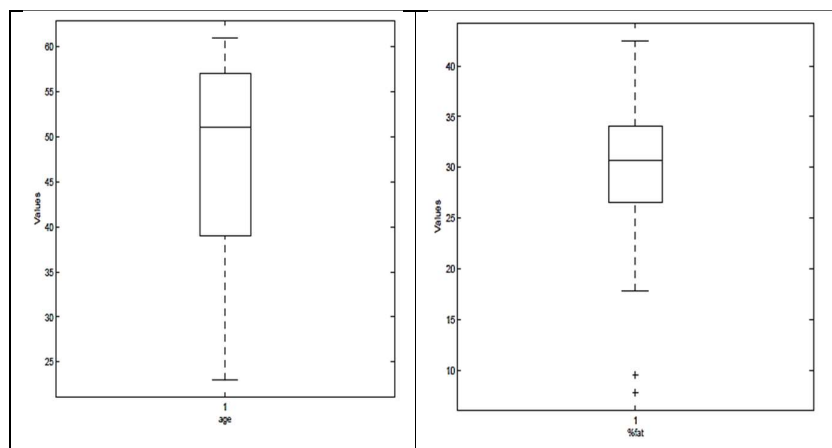


- Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
- Find the mean and median
 - What is the mode of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.)
 - What is the midrange of the data?
 - Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data.
 - Give the five-number summary of the data.
- The (arithmetic) mean of the data is $= 809/27 = 30$. The median (middle value of the ordered set, as the number of values in the set is odd) of the data is: 25.
 - This data set has two values that occur with the same highest frequency and is, therefore, bimodal. The modes (values occurring with the greatest frequency) of the data are 25 and 35.
 - The midrange (average of the largest and smallest values in the data set) of the data is: $(70+13)/2 = 41.5$.
 - The first quartile (corresponding to the 25th percentile) of the data is: 20. The third quartile (corresponding to the 75th percentile) of the data is: 35.
 - The five number summary of a distribution consists of the minimum value, first quartile, median value, third quartile, and maximum value. It provides a good summary of the shape of the distribution and for this data is: 13, 20, 25, 35, 70.

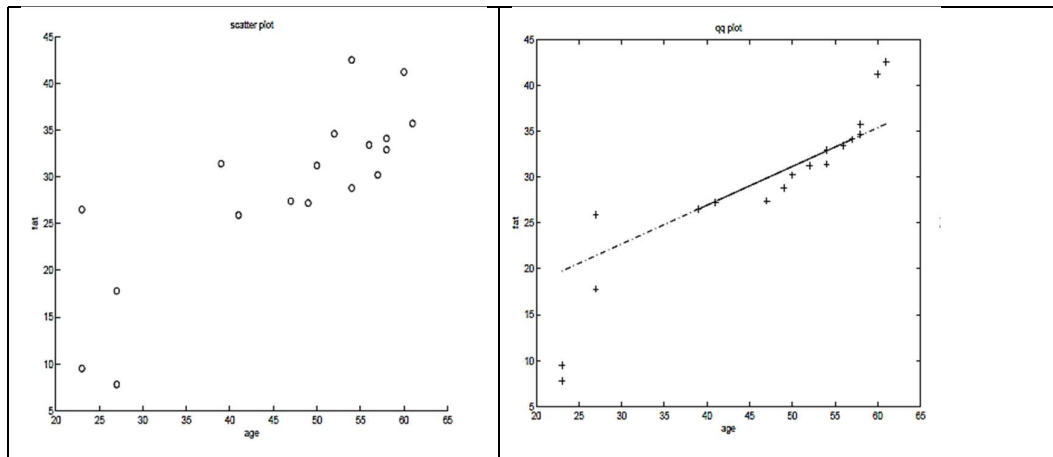
- Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result:

Age	23	23	27	27	39	41	47	49	50
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
Age	52	54	54	56	57	58	58	60	61
% fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- Calculate the mean, median, and standard deviation of age and %fat
 - Draw the boxplots for age and %fat
 - Draw a scatter plot and a q-q plot based on these two variables
 - Calculate the correlation coefficient (Pearson's product moment coefficient). Are these two variables positively or negatively correlated?
- For the variable age the mean= 46.44, the median =51, and standard deviation=12.85.
- For the variable %fat the mean= 28.78, the median=30.7, and the standard deviation=8.99.
- Draw the boxplots for age and %fat



iii. Draw a scatter plot and a q-q plot based on these two variables



iv. Calculate the correlation coefficient (Pearson's product moment coefficient). Are these two variables positively or negatively correlated?

The Pearson correlation coefficient is 0.82, the variables are positively correlated.

1.7 Data Visualization

- Data visualization aims to communicate data clearly and effectively through graphical representation.
 - To discover data relationships that are otherwise not easily observable by looking at the raw data.
 - To create fun and interesting graphics.

Several representative approaches are:

1.7.1 Pixel-Oriented Visualization Techniques

1.7.2 Geometric Projection Visualization Techniques

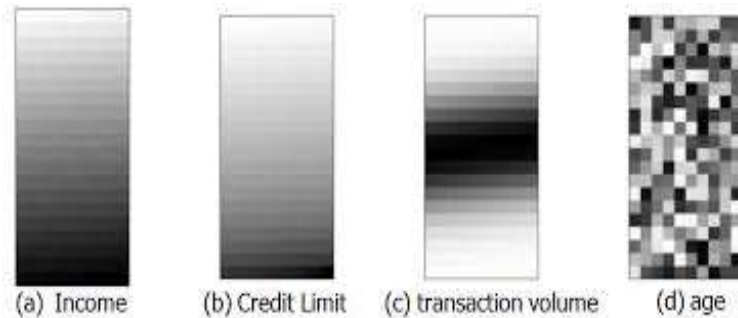
1.7.3 Icon-Based Visualization Techniques

1.7.4 Hierarchical Visualization Techniques

1.7.5 Visualizing Complex Data and Relations

1.7.1 Pixel-Oriented Visualization Techniques

- A simple way to visualize the value of a dimension is to use a pixel where the color of the pixel reflects the dimension's value.
- For a data set of m dimensions, create m windows on the screen, one for each dimension.
- The m dimension values of a record are mapped to m pixels at the corresponding positions in the windows.
- The colors of the pixels reflect the corresponding values.
- Inside a window, the data values are arranged in some global order shared by all windows. The global order may be obtained by sorting all data records in a way that's meaningful for the task at hand.
- In pixel-oriented techniques, data records can also be ordered in a query-dependent way.



Pixel-oriented visualization of four attributes by sorting all customers in income ascending order.

- The first pixel in a row is far away from the last pixel in the previous row, though they are next to each other in the global order. Moreover, a pixel is next to the one above it in the window, even though the two are not next to each other in the global order.

1.7.2 Geometric Projection Visualization Techniques

- A drawback of pixel-oriented visualization techniques is that they cannot help us much in understanding the distribution of data (do not show whether there is a dense area) in a multidimensional space.
- Geometric projection techniques help users find interesting projections of multidimensional data sets.
- A 3-D scatter plot uses three axes in a Cartesian coordinate system. If it also uses color, it can display up to 4-D data points.
- For an n dimensional data set, a scatter-plot matrix is an $n \times n$ grid of 2-D scatter plots that provides a visualization of each dimension with every other dimension.

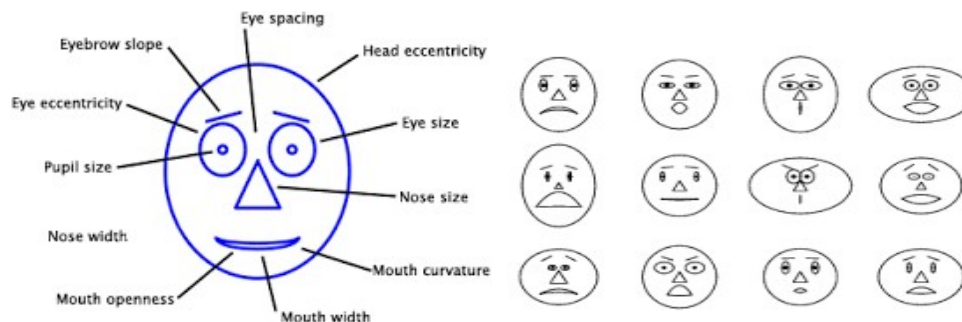
1.7.3 Icon-Based Visualization Techniques

- Icon-based visualization techniques use small icons to represent multidimensional data values. Two popular icon-based techniques:

- 1) Chernoff faces and
- 2) Stick figures.

1) **Chernoff faces** (1973 Herman Chernoff) display multidimensional data of up to 18 variables (or dimensions) as a cartoon human face (Fig).

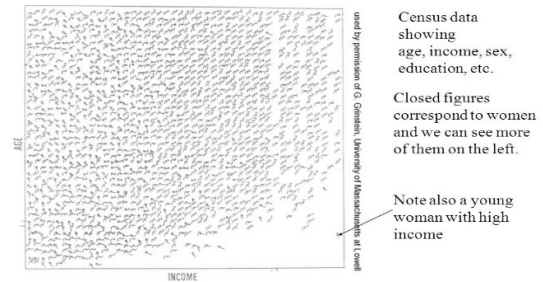
- They help reveal trends in the data. Components of the face, such as the eyes, ears, mouth, and nose, represent values of the dimensions by their shape, size, placement, and orientation.
- They make use of the ability of the human mind to recognize small differences in facial characteristics and to assimilate many facial characteristics at once.
- They facilitate visualization of regularities and irregularities present in the data.



2) The **Stick figure visualization technique** maps multidimensional data to five-piece stick figures, where each figure has four limbs and a body.

- Two dimensions are mapped to the display (x and y) axes and the remaining dimensions are mapped to the angle and/or length of the limbs.
- Fig shows census data, where age and income are mapped to the display axes, and the remaining dimensions (gender, education, and so on) are mapped to stick figures.
- If the data items are relatively dense with respect to the two display dimensions, the resulting visualization shows texture patterns, reflecting data trends.

Illustration of Stick Figure Use



1.7.4 Hierarchical Visualization Techniques

- Hierarchical visualization techniques partition all dimensions into subsets (i.e., subspaces). The subspaces are visualized in a hierarchical manner.
- “Worlds-within-Worlds,” also known as n-Vision, is a representative hierarchical visualization method. A user can interactively change, in the outer world, the location of the origin of the inner world. The user then views the resulting changes of the inner world.
- Tree-maps display hierarchical data as a set of nested rectangles. For example, Fig shows a tree-map visualizing Google news stories. All news stories are organized into seven categories, each shown in a large rectangle of a unique color. Within each category (i.e., each rectangle at the top level), the news stories are further partitioned into smaller subcategories.

1.7.5 Visualizing Complex Data and Relations

- Hierarchical visualization techniques partition all dimensions into subsets (i.e., subspaces). The subspaces are visualized in a hierarchical manner.
- A Tag Cloud is a visualization of statistics of user-generated tags in which, tags are listed alphabetically or in a user-preferred order.
- The importance of a tag is indicated by font size or color.
- Figure 1.8.5 shows a tag cloud for visualizing the popular tags used in a Website.
- Tag clouds are often used in two ways. First, in a tag cloud for a single item, we can use the size of a tag to represent the number of times that the tag is applied to this item by different users.
- Second, when visualizing the tag statistics on multiple items, we can use the size of a tag to represent the number of items that the tag has been applied to, that is, the popularity of the tag.

1.8 Dissimilarity of Numeric Data on Distance measures: Euclidean, Manhattan, and Minkowski

Distance measures are commonly used for computing the dissimilarity of objects described by numeric attributes.

These measures include the

1) Euclidean distance

2) Manhattan distance

3) Minkowski distance

Let $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ be two objects described by p numeric attributes.

The Euclidean distance between objects i and j is defined as:

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{in} - x_{jn})^2}.$$

The Manhattan (or city block) distance, named so because it is the distance in blocks between any two points in a city (such as 2 blocks down and 3 blocks over for a total of 5 blocks). It is defined as:

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{in} - x_{jn}|$$

The Minkowski distance is a generalization of the Euclidean and Manhattan distances. It is defined

as:
$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h}$$

Given two objects represented by the tuples (22,1,42,10) and (20,0,36,8):

- Compute the Euclidean distance between the two objects.
- Compute the Manhattan distance between the two objects.
- Compute the Minkowski distance between the two objects, using $h = 3$.

a) Euclidean distance between Numeric attributes:

$$\begin{aligned} d(i, j) &= \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{in} - x_{jn})^2} \\ &= \sqrt{(22 - 20)^2 + (1 - 0)^2 + (42 - 36)^2 + (10 - 8)^2} \\ &= \sqrt{45} = 6.7082 \end{aligned}$$

b) Manhattan distance between Numeric attributes:

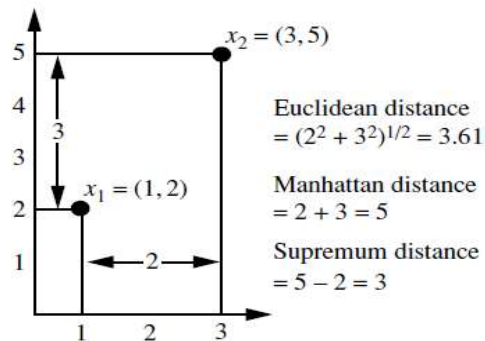
$$\begin{aligned} d(i, j) &= |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{in} - x_{jn}| \\ &= |22 - 20| + |1 - 0| + |42 - 36| + |10 - 8| \\ &= 11 \end{aligned}$$

c) Minkowski distance between Numeric attributes:

$$\begin{aligned} d(i, j) &= \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h} \\ &= \sqrt[3]{|22 - 20|^3 + |1 - 0|^3 + |42 - 36|^3 + |10 - 8|^3} = \sqrt[3]{233} \\ &= 6.1534 \end{aligned}$$

- Let $x_1 = (1, 2)$ and $x_2 = (3, 5)$ represent two objects as shown in Figure.

- The Euclidean distance between the two is $\sqrt{2^2 + 3^2} = 3.61$



- The Manhattan distance between the two is $2+3 = 5$

DATA PREPROCESSING

Data Quality: Why Preprocess the Data?

Data Preprocessing is the process of making the data more suitable for data mining.

- Data have quality if they satisfy the requirements of the intended use. There are many factors comprising data quality, including accuracy, completeness, consistency, timeliness, believability, and interpretability.
- Data in the real world is:
 - ✓ incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - ✓ inaccurate (noisy): containing errors or outliers
 - ✓ inconsistent: containing discrepancies in codes or names
- No quality data, no quality mining results!
- ✓ Quality decisions must be based on quality data
- ✓ Data warehouse needs consistent integration of quality data

1.9 Major Tasks (forms) in Data Preprocessing

- Data cleaning: Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies.
- Data integration: Integration of multiple databases, data cubes, or files.
- Data reduction: Obtains reduced representation in volume but produces the same or similar analytical results.
 - Dimensionality reduction: reduced or “compressed” representation of the original data
 - Numerosity reduction: replaced by alternative, smaller representations using parametric models or nonparametric models
- Data transformation: Normalization and aggregation.
- Data Discretization and concept hierarchy generation: Powerful tools for data mining at multiple abstraction levels. Normalization, data discretization, and concept hierarchy generation are forms of data transformation

1.10 Data Cleaning

- Data cleaning tasks
 - Fill in MISSING VALUES
 - Identify outliers and smooth out NOISY DATA
 - Correct inconsistent data

Missing Data

- Data is not always available
- ✓ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- **Missing data may be due to**
 - ✓ equipment malfunction
 - ✓ data not entered due to misunderstanding
 - ✓ inconsistent with other recorded data and thus deleted
 - ✓ certain data may not be considered important at the time of entry
 - ✓ not register history or changes of the data
- Missing data may need to be inferred.
- **How to Handle Missing Data?**
 - Ignore the tuple: usually done when class label is missing (assuming the tasks in classification—not effective when the percentage of missing values per attribute varies considerably).
 - Fill in the missing value manually: tedious + infeasible
 - Use a global constant to fill in the missing value: e.g., “unknown”, a new class?!
 - Use the attribute mean to fill in the missing value
 - Use the attribute mean for all samples belonging to the same class to fill in the missing value: smarter
 - Use the most probable value to fill in the missing value: inference-based such as Bayesian formula or decision tree

Noisy Data

- Noise: random error or variance in a measured variable
- **Incorrect attribute values may due to**
 1. faulty data collection instruments
 2. data entry problems
 3. data transmission problems
 4. technology limitation
 5. inconsistency in naming convention
- **How to Handle Noisy Data?**
 1. Binning method: first sort data and partition into (equi-depth) bins and then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
 2. Regression: smooth by fitting the data into regression functions
 3. Clustering: detect and remove outliers
 4. Combined computer and human inspection: detect suspicious values and check by human

1. Binning Method

- Binning methods smooth a sorted data value by consulting its “neighborhood,” that is, the values around it. The sorted values are distributed into a number of “buckets,” or bins.

1) equal- depth (frequency) partitioning

2) equal-width (distance) partitioning

3) Clustering

1) Equal-width partitioning:

- It divides the range into N intervals of equal size: uniform grid
- If A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$
- The width of each interval will be $(215 - 5)/3 = 70$.
 - Bin 1: 5, 10, 11, 13, 15, 35, 50, 55, 72, i.e. all values from 5 and 75
 - Bin 2: 92 i.e. all values from 76 and 145
 - Bin 3: 204,215 i.e. all values from 146 and 215

➤ Equal-depth (frequency) partitioning:

- It divides the range into N intervals, each containing approximately same number of samples
- Good data scaling but, Managing categorical attributes can be tricky.

Ex: Random numbers: 21, 24, 4, 9, 8, 15, 21, 34, 28, 29, 26, 25

Sorted data: 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

1)Partition into frequency bins:

Bins of size 4 (i.e., each bin contains four values)

- Bin 1: 4, 8, 9, 15
- Bin 2: 21, 21, 24, 25
- Bin 3: 26, 28, 29, 34

2)Smoothing by bin means:

Each value in a bin is replaced by the mean value of the bin

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

3)Smoothing by bin boundaries:

The min and max values in a given bin are identified as the bin boundaries and replaced by the closest boundary value.

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

Binning Example

- Suppose a group of 12 sales price records has been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods.

a) equal-frequency (equi-depth) partitioning

- Bin 1: 5, 10, 11, 13
- Bin 2: 5, 35, 50, 55
- Bin 3: 72, 92, 204, 215

b) equal-width partitioning

- The width of each interval will be $\text{Width} = (\text{Max}-\text{Min})/\text{No.of Bins}$
- $(215 - 5)/3 = 70$.
 - Bin 1: 5, 10, 11, 13, 15, 35, 50, 55, 72, i.e. all values from 5 and 75
 - Bin 2: 92 i.e. all values from 76 and 145
 - Bin 3: 204,215 i.e. all values from 146 and 215

c) clustering

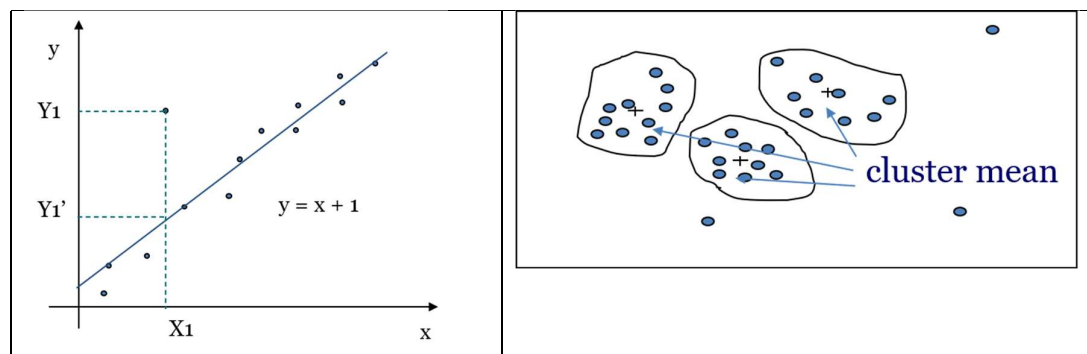
We will use a simple clustering technique, divide the data along the 2 biggest gaps in the data.

- Bin 1: 5, 10, 11, 13, 15
- Bin 2: 35, 50, 55, 72, 92
- Bin 3: 204, 215

2. Regression

- Data can be smoothed by fitting the data to a function, with Regression
- Linear regression involves finding the “best” line to fit two attributes (or variables), so that one attribute can be used to predict the other.
- Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

3. Clustering: Outliers may be detected by clustering, where similar values are organized into groups, or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered outliers.



Data Cleaning as a Process

- The first step in data cleaning as a process is discrepancy detection.
 - 1) Discrepancies (Inconsistencies) can be caused by several factors:
 - poorly designed data entry forms that have many optional fields
 - human error in data entry
 - deliberate errors (e.g., respondents not wanting to divulge information about themselves like Income, Contact Nos)
 - data decay (e.g., outdated addresses).
 - inconsistent data representations and codes. (Ex: No. Num #)
 - errors in devices that record data and system errors
 - given attribute can have different names in different databases. (Regd No. $\equiv$ Roll No., Gender $\equiv$ Sex)
 - 2) Field overloading when developers squeeze new attribute definitions into unused (bit) portions of already defined attributes (e.g., an unused bit of an attribute that has a value range that uses only, say, 31 out of 32 bits).The data should also be examined regarding unique rules, consecutive rules, and null rules.
- A **Unique rule** says that each value of the given attribute must be different from all other values for that attribute.
- A **Consecutive rule** says that there can be no missing values between the lowest and highest values for the attribute, and that all values must also be unique (e.g., as in check numbers).
- A **Null rule** specifies the use of blanks, question marks(?), special characters, or other strings that may indicate the null condition (e.g., where a value for a given attribute is not available). There are a number of different commercial tools that can aid in the discrepancy detection step.
- **Data scrubbing tools** use simple domain knowledge (e.g., knowledge of postal addresses and spell-checking) to detect errors and make corrections in the data. These tools rely on parsing and fuzzy matching techniques when cleaning data from multiple sources.

- **Data auditing tools** find discrepancies by analyzing the data to discover rules and relationships, and detecting data that violate such conditions.
- New approaches to data cleaning emphasize increased interactivity.

1.11 Data Integration

- Data integration is the process of *integrating (merging) data from multiple data stores into a coherent store.*
 - *to probably have a single view over all these sources*
 - *and answering queries using the combined information*
- Integration can be:
 - Physical: Copying the data to warehouse
 - Virtual: Keep the data only at the sources
- Data integration is also valid within a single organization integrating data from different departments or sectors
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies in the resulting data set and improve mining speed and quality.
- The semantic heterogeneity and structure of data pose great challenges in data integration.
- Reasons of Mismatching
 - Misspelling: “Smith”, “Smeth”, “Snith”
 - Variant names, synonyms, and abbreviations:
 - “St.”, “St”, “Street”; “Prof”, “Professor”; “car”, “vehicle”
- 3) Different systems:
 - “Chin Le”, “Le, Chin”... “10/02/2000”, “10-02-2000”, “02-10-2000”
- 4) Different domains: “YES/NO”; “Y/N”; “1/0”; “T/F”

1) Entity identification problem

During data integration like *Schema integration* and *object matching* from one database to another, special attention must be paid to the structure of the data.

- This is to ensure that any attribute functional dependencies and referential constraints in the source system match those in the target system.
- Detecting and resolving data value conflicts:
 - for the same real world entity, attribute values from different sources are different.
 - Ex: cust-id ≡ cust-# ≡ cust-no
- different representations, different scales.
 - Ex: Metric vs. British (Imperial) units, Sqm vs Sft, Celsius vs Fahrenheit, grams vs pounds

Handling Redundant Data in Data Integration

- 1) Redundant data occur often when integration of multiple databases
 - The same attribute may have different names in different databases.
 - One attribute may be a “derived” attribute in another table, e.g., annual revenue
- 2) Redundant data may be able to be detected by correlational analysis.
 - For nominal data -> χ^2 (chi-square) test.
 - For numeric attributes -> correlation coefficient and covariance.
- 3) Duplication should also be detected at the tuple level.
- 4) The use of denormalized tables (avoiding joins) is another source of data redundancy.

X² Correlation Test for Nominal (Categorical) Data

- X² (chi-square) test (Pearson X² statistic)
- The larger the X² value, the more likely the variables are related
- The cells that contribute the most to the X² value are those whose actual count is very different from the expected count represented in parentheses.
- Chi-Square Calculation: An Example 2 x 2 Contingency Table Data for “are gender and preferred reading correlated”?

	male	female	Sum (row)
fiction	250 (90)	200 (360)	450
non_fiction	50 (210)	1000 (840)	1050
Sum(col.)	300	1200	1500

$$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840}$$
$$= 284.44 + 121.90 + 71.11 + 30.48 = 507.93$$

- Statistically, significance level is 10.828. Since our computed value is above this, we can conclude that the two attributes *gender* and *preferred reading* are (strongly) correlated for the given group of people.

1.12 Data Reduction

- Data reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet produces the same (or almost the same) analytical results.

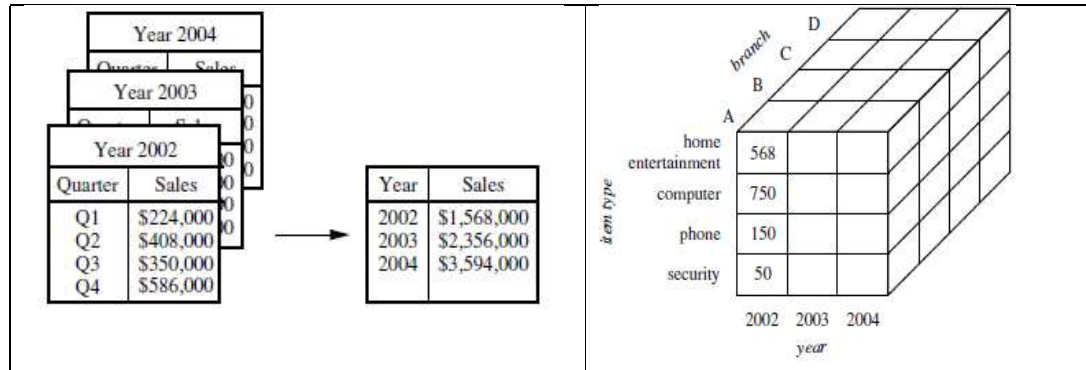
Data reduction strategies:

- 1.12.1. Data cube aggregation: applied to the data in the construction of a data cube.
- 1.12.2. Attribute subset selection: irrelevant, weakly relevant, or redundant attributes or dimensions may be detected and removed.
- 1.12.3. Dimensionality reduction: encoding mechanisms are used to reduce the data set size.
- 1.12.4. Numerosity reduction: where the data are replaced or estimated by alternative, smaller data representations such as parametric models or nonparametric methods such as clustering, sampling, and histograms.
- 1.12.5. Discretization and concept hierarchy generation: where raw data values for attributes are replaced by ranges or higher conceptual levels. Very useful for the automatic generation of concept hierarchies and allow the mining of data at multiple levels of abstraction.

1.12.1 Data Cube Aggregation:

- The data can be aggregated so that the resulting data is summarized, smaller in volume, without loss of information necessary for the analysis task. Ex: total sales per year instead of per quarter
- Data cubes store multidimensional aggregated information. Each cell holds an aggregate data value, corresponding to the data point in multidimensional space.
- Concept hierarchies may exist for each attribute, allowing the analysis of data at multiple levels of abstraction. Data cubes provide fast access to precomputed, summarized data, thereby benefiting OLAP as well as data mining.
- The cube created at the lowest level of abstraction is referred to as the base cuboid, individual entity of interest, such as sales or customer. A cube at the highest level of abstraction is the

apex cuboid. Data cubes created for varying levels of abstraction are often referred to as cuboids, so that a data cube may instead refer to a lattice of cuboids.



1.12.2 Attribute subset selection:

- Heuristic methods of attribute subset selection include the following techniques:
- 1) **Stepwise forward selection:** The procedure starts with an empty set of attributes as the reduced set. The best of the original attributes is determined and added to the reduced set. At each subsequent iteration or step, the best of the remaining original attributes is added to the set.
- 2) **Stepwise backward elimination:** The procedure starts with the full set of attributes. At each step, it removes the worst attribute remaining in the set.
- 3) **Combination of forward selection and backward elimination:** The stepwise forward selection and backward elimination methods can be combined so that, at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

1.12.3 Dimensionality reduction

- Data encoding or transformations are applied so as to obtain a reduced or “compressed” representation of the original data.
- If the original data can be reconstructed from the compressed data without any loss of information, the data reduction is called lossless.
- If we can reconstruct only an approximation of the original data, then the data reduction is called lossy.
- There are several well-tuned algorithms for string compression. Although they are typically lossless, they allow only limited manipulation of the data.
- Two popular and effective methods of lossy dimensionality reduction:
 - 1) wavelet transforms and
 - 2) principal components analysis.

1.12.4 Numerosity Reduction

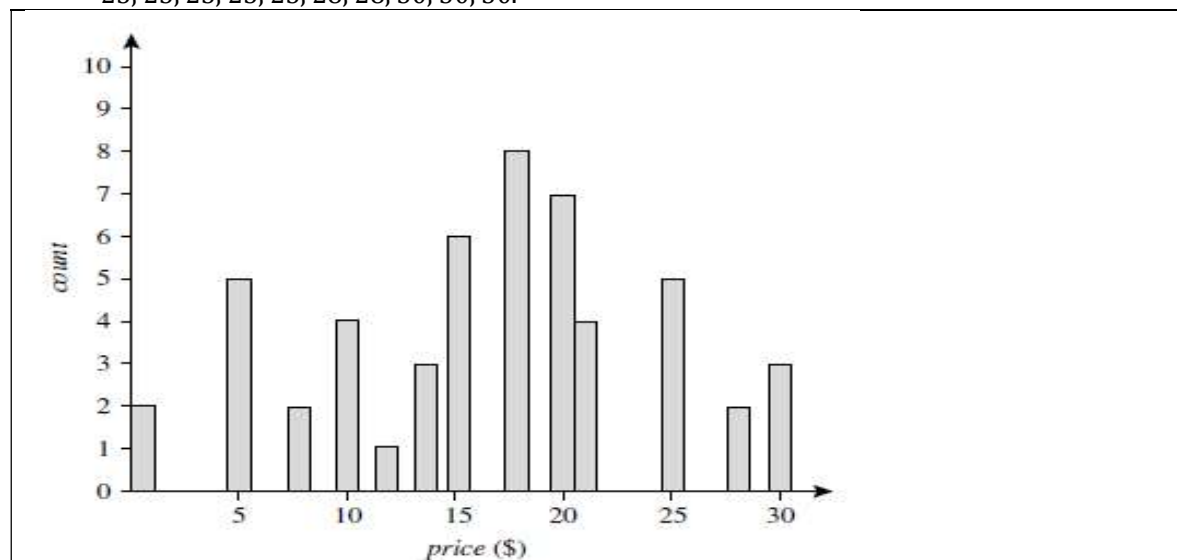
- Techniques of numerosity reduction can indeed be applied for reducing the data volume by choosing alternative, 'smaller' forms of data representation.
- These techniques may be 1) parametric or 2) nonparametric

1) Parametric methods, a model is used to estimate the data, so that typically only the data parameters need to be stored, instead of the actual data. (Outliers may also be stored.)

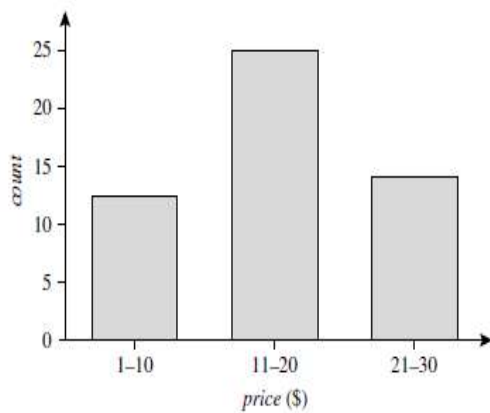
Ex: Log-linear models, which estimate discrete multidimensional probability distributions, are an example.

2) Nonparametric methods for storing reduced representations of the data (do not assume models) include histograms, clustering, and sampling.

- Linear regression: Data are modeled to fit a straight line, often uses the least-square method to fit the line. $Y = \alpha + \beta X$
 - Two parameters, α and β specify the line and are to be estimated by using the data at hand.
 - using the least squares criterion to the known values of $Y_1, Y_2, \dots, X_1, X_2, \dots$
- Multiple regression: allows a response variable Y to be modeled as a linear function of multidimensional feature vector.
 - $Y = b_0 + b_1 X_1 + b_2 X_2$.
 - Many nonlinear functions can be transformed into the above.
- Log-linear model: approximates discrete multidimensional probability distributions
 - The multi-way table of joint probabilities is approximated by a product of lower-order tables.
 - Probability: $p(a, b, c, d) = \alpha ab \beta ac \gamma ad \delta bcd$
- Histograms use binning to approximate data distributions and are a popular form of data reduction.
 - Divide data into buckets and store average (sum) for each bucket
 - Can be constructed optimally in one dimension using dynamic programming
 - Related to quantization problems.
 - The following data are a list of AllElectronics prices for commonly sold items (rounded to the nearest dollar). The numbers have been sorted: 1, 1, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.



A histogram for price using singleton buckets—each bucket represents one price-value/frequency pair.



An equal-width histogram for price, where values are aggregated so that each bucket has a uniform width of \$10.

1.13 Data Transformation and Data Discretization

- In data transformation, the data are transformed or consolidated into forms appropriate for mining.
- Strategies for data transformation include the following:
 - 1) **Smoothing**: remove noise from data. Binning, regression, and clustering.
 - 2) **Aggregation**: summarization, data cube construction
 - 3) **Generalization**: concept hierarchy climbing like street to city or country
 - 4) **Normalization**: scaled to fall within a small, specified range, such as -1.0 to 1.0, or 0.0 to 1.0. min-max normalization, z-score normalization, normalization by decimal scaling
 - 5) **Attribute/feature construction** New attributes constructed from the given ones.
 - 6) **Discretization**: where the raw values of a numeric attribute (e.g., age) are replaced by interval labels (e.g., 0-10, 11-20, etc.) or conceptual labels (e.g., youth, adult, senior).

Data Transformation: Normalization

1) Min-max normalization: performs a linear transformation on the original data. Suppose that $\min_A$ and $\max_A$ are the minimum and maximum values of an attribute, A.

- Suppose that $\min_A$ and $\max_A$ are the minimum and maximum values of an attribute, A. Min-max normalization maps a value, v_i , of A to v'_i in the range $[\text{new_min}_A, \text{new_max}_A]$ by computing.

$$v'_i = \frac{v_i - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

- Min-max normalization preserves the relationships among the original data values.
- It will encounter an “out-of-bounds” error if a future input case for normalization falls outside of the original data range for A.
- Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. The \$73,600 is transformed to

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

2) z-score normalization: (or zero-mean normalization), the values for an attribute, A, are normalized based on the mean (i.e., average) and standard deviation of A. A value, v_i of A is normalized to v'_i by computing

$$v'_i = \frac{v_i - \bar{A}}{\sigma_A},$$

$\bar{A}$ = mean = $(v_1 + v_2 + \dots + v_n)/n$
 σ_A = standard deviation

- This method of normalization is useful when the actual minimum and maximum of attribute A are unknown, or when there are outliers that dominate the min-max normalization.

Ex: Suppose that the mean and standard deviation of the values for the attribute income are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for income is transformed to:

$$\frac{73,600 - 54,000}{16,000} = 1.225$$

- A variation of this z-score normalization replaces the standard deviation equation by the mean absolute deviation of A, denoted s_A , is

$$s_A = \frac{1}{n}(|v_1 - \bar{A}| + |v_2 - \bar{A}| + \dots + |v_n - \bar{A}|)$$

- Thus, z-score normalization using the mean absolute deviation is
- The mean absolute deviation, s_A , is more robust to outliers than the standard deviation,

$$v'_i = \frac{v_i - \bar{A}}{s_A}$$

$$v' = \frac{v}{10^j}$$

3) Normalization by decimal scaling is given by: j is $\text{Max}(|v'|) < 1$

Value ranges

a) min-max normalization: range is [new_min, new_max]

b) z-score normalization: range is [(old_min-mean)/stddev, (old_max-mean)/stddev]. range for all possible data sets is $(-\infty, +\infty)$.

c) normalization by decimal scaling: range is $(-1.0, 1.0)$.

Example: The age values for the data tuples are (in increasing order) 12, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 72.

i) Use min-max normalization to transform the value 35 for age onto the range [0.0, 1.0].

Using the corresponding equation with $\text{min}A = 13$, $\text{max}A = 70$, new $\text{min}A = 0.0$, new $\text{max}A = 1.0$, then $v = 35$ is transformed to $v' = 0.39$.

ii) And use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.

Using the corresponding equation where $\bar{A} = 809/27 = 29.96$ and $\sigma_A = 12.94$, then $v = 35$ is transformed to $v' = 0.39$.

iii) Use normalization by decimal scaling to transform the value 35 for age.

Using the corresponding equation where $j = 2$, $v = 35$ is transformed to $v' = 0.35$.

Discretization

Typical methods: All the methods can be applied recursively

1) Discretization by Binning

- Top-down split, unsupervised,

2) Discretization by Histogram Analysis

- Top-down split, unsupervised

3) Discretization by Cluster, Decision Tree, and Correlation Analyses

- Either top-down split or bottom-up merge, unsupervised
- Entropy-based discretization: supervised, top-down split
- Interval merging by X^2 Analysis: unsupervised, bottom-up merge
- Segmentation by natural partitioning: top-down split, unsupervised

*_*_*